OPTIONS AGENT · ALPACA HACKATHON

Three books, one pipeline.

A short-volatility carry book, a long-gamma intraday book and an LLM-driven earnings book — run through one gated decision pipeline in which no model is allowed to approve a trade.

+$5,134	1.03	−3.28%	5.1%
carry book · 160 trades · post-audit	profit factor, both books	max drawdown	of ideas approved

Real Alpaca bars, Feb–Aug 2026 · 409 closed trades after the 30 Aug credit-to-width gate · spread and fees charged inside every fill. The pre-gate +13.8% headline was ~93% artefact and is shown, not hidden, on the EVIDENCE slide. The events book has no usable backtest and says so.

THE THESIS

Three bets that fail in different weather

Uncorrelated in signal, opposite in volatility exposure, and only one of them needs a market condition to occur. A dry week for one is not a dry week for the account.

CARRY

`vol_carry`

Options are priced for more movement than actually happens.

Bet	Sell movement, paid up front
Shape	Iron condor, 25Δ, 7–14 DTE
Wants	Nothing to happen
Hold	3–10 sessions, overnight
Edge	The IV − RV spread
Risk	Short gamma

INTRADAY

`intraday_momentum`

A move with a mechanism behind it keeps going for a while.

Bet	Buy the continuation
Shape	Long 0–2 DTE, or a vertical
Wants	Something to happen, fast
Hold	Minutes, flat by 15:10
Edge	Directional continuation
Risk	Spread cost, signal decay

EVENTS

`earnings_event_directional`

A scheduled print is an event the option market has to price.

Bet	Trade the mispriced move
Shape	Condor if rich, vertical if cheap
Wants	The priced move to be wrong
Hold	One night, flat by 09:45
Edge	Implied vs realised reaction
Risk	The gap — bounded by structure

Carry and intraday share one pool of capital, so a firewall — not politeness — keeps them apart. The events book leases none of it and is bounded by size instead.

Four hard gates, then economics

Hard gates, not a blended score. A score would let a rich IV reading paper over an earnings date — the one trade this book must never take.

1 Premium

`IV rank ≥ 0.35  AND  IV − RV ≥ 3 vol points`

Rank says vol is high for this name. The spread says you are paid more than it has actually been moving. The spread is the edge; the rank is only a regime filter.

2 Trend

`ADX ≤ 25  and  |trend strength| ≤ 0.60`

Short premium is short movement. A name that is going somewhere is disqualified before anything else is measured.

3 Event

`No earnings in the expiry window · no ex-div under a short call`

IV before earnings is elevated because of the event. Selling it is selling a coin flip at fair odds.

4 Macro lens

`Sector-wide vol → sell.  Idiosyncratic → veto.`

A shared move has no single headline behind it. A name-specific one does, and a short condor has no defence against the next one.

Then the cost gate: **round-trip spread ≤ 20% of the credit received** — a four-leg condor crosses eight half-spreads on a round trip, so this is most of the P&L, not a nicety.

The exit pair sets the hit rate you need

Breakeven hit rate = stop ÷ (target + stop). It is arithmetic, and it is decided before any trade is taken.

Old: 30% target / 2.0× stop

87.0% needed, before any spread

Observed win rate was 88%. One point of margin — and the 20% cost ceiling implied 95.7%, so the gate was admitting trades the arithmetic said would lose.

Now: 50% target / 1.5× stop

75.0% needed, before any spread

The same 20% cost ceiling now implies 85% — three points of margin. Same gate, opposite meaning, because the exits changed underneath it.

Checked in this order, every cycle

$450 hard dollar stop
checked first

50% of max profit
close

loss = 1.5× credit
close

3 DTE floor
close regardless of P&L

short strike touched
close the tested side

macro flag / flatten
close at next open

The dollar stop was $900 and never fired: the six largest losses were $780–868, all sitting just underneath it, and every one exited on the DTE floor instead — which realises whatever has accumulated rather than capping it. A stop set above the loss distribution is decoration.

Three indicators, three different questions

Only VWAP has an opinion on direction, so the other two cannot quietly agree with it and call that confluence.

VWAP
TRIGGER

Session VWAP cross, or price on the band within 2.0× the session's usual dispersion. Searched over the last 15 one-minute bars — exactly one polling gap, so a cross cannot fall between two cycles unseen.

Bollinger width
FILTER

Width, not position. Position is a mean-reversion signal and would fight VWAP on every candidate. Width asks one thing: is the volatility regime expanding?

RSI
VETO ONLY

One-sided, at 80/20 rather than 70/30. Strong momentum carries elevated RSI for long stretches, so a tight threshold rejects exactly the trades this book exists to take.

The non-generic layer

A VWAP cross with no reason behind it is drift, and drift reverts. A cross with a mechanism persists.

Catalyst score

News	0.50
Breadth	0.35
Volume participation	0.15

Rank-normalised within each source before the weights are applied — raw units across sources are not comparable. Deterministic by default: an LLM call inside a loop whose signal decays in minutes is a latency risk.

Hard, always: the spread gate

`(ask − bid) / mid ≤ 4%   and   round-trip spread ≤ 30% of target`

Target profit is $10–30 on a $2.00 contract. A $0.10-wide single-name quote costs $20 round trip — the whole target. Index ETFs only; the ceiling was 2% until measurement showed the tightest quote in the universe was 2.84%, so the gate was rejecting the market rather than selecting within it.

Eight vetoes in a row is not a strategy

Each gate passed roughly 70% of candidates. Stacked as hard vetoes, that is $0.7^8 \approx 6\%$ — a book arithmetically designed almost never to fire.

Measured over 864 candidates, 9 sessions

424 died on the volume gate alone

0 survived the whole chain. Loosening any one gate simply promoted the next one to being the wall.

The fix: two hard gates, everything else votes

VWAP trigger	stays HARD — it decides direction. Without it there is no trade to make.
Spread gate	stays HARD — it is economics, not evidence.
Everything else	votes. Three of the available votes must agree.

The seven voters — each asks a question the others cannot answer

Volume z ≥ 0.5
in the same time-of-day bucket

Persistence
2 bars — not a single-bar spike

Band width rising
is the regime expanding?

Higher timeframe
is the hour with us or against?

Term structure
does the surface expect anything?

RSI
are we buying exhaustion?

Catalyst
is there a mechanism?

“Unmeasurable” is not “failed”: needed = min(3, possible), so a vote that cannot be evaluated raises neither side and costs a candidate nothing.

The only book whose gate opens on a date

The other two wait for a market condition to cross a threshold, and the recurring failure is that it never crosses. Broadcom reports on 2 September whether or not any indicator agrees.

What is measured

```
implied move ÷ median |reaction|, last 4 prints
```

That is a volatility reading — how far the option market's priced move for this print sits from what the name actually does when it reports. It is a ranking device, not an edge estimate: four quarters per name is not a sample you can walk forward.

Screen gates

Implied move ≥ 3%	below that, no event is priced in
Relative spread ≤ 25%	THE binding gate
Expiry contains the print	the most expensive mistake available
Top 10 names	one night's risk budget, shared

The model proposes. A file confirms.

Featherless serves open-weight models whose weights were frozen months ago. Asked to list next week's reporters, one answers fluently and sometimes wrongly — and a wrong date is not a bad trade, it is a position opened against no event at all. So the model only proposes candidates; config/events/earnings_calendar.json confirms every one of them. Anything unverified is logged and never armed.

CPRT

ships with confirmed: false because Copart had not announced its date. It will not trade.

Sizing is bounded three ways instead of by the firewall: 1.2% per name · 4% shared across everything opened that night · 10 contracts absolute, a ceiling that does not scale with equity.

Read the name for days, not for one minute

The information that decides a print does not arrive in a single window. A snapshot sees the last item on the wire and calls it the picture.

1

Poll

Hourly, 04:00–16:00 ET, from three days before a confirmed report. Alpaca news plus StockTwits.

2

Deduplicate

Every item already read is discarded — keyed by a hash of its timestamp and text.

3

Nothing new?

Stop. No model call, no note, no tokens. A quiet name is a real state and is reported as one.

4

Judge the new items only

One narrow question: is this material to how the stock trades on the print, and which way does it point?

5

Keep or drop

Kept as a dated note if salience ≥ 0.35. Aged out at 10 days, capped at 40.

It stops at the print. The day after a name reports its dossier is retired — post-print commentary left in a live dossier would be read by a later cycle as pre-print evidence, which is the most expensive kind of quiet error this book can make.

Two roles, two models

watch triage	~40 calls/day	`Qwen3-8B`
direction call	once per name	`Qwen3-32B`

The triage is narrow, schema-bound and usually right to say "immaterial". Running the big model on it spends the direction call's budget on noise.

What bounds the model

Confidence floor 0.55	and it sets the size, so a marginal call trades small
Evidence required	a confident call citing nothing is a guess with a number attached
Abstention rate journalled	zero abstentions logs a warning — a model that never declines is not filtering
Evidence is untrusted text	fenced, stripped, truncated, parsed to a fixed schema, every field clamped

We measured volatility and traded direction

A directional structure bought at a fair-to-rich implied move, on a direction call no better than a coin flip, returns minus the round trip in expectation. That is arithmetic, not a sample-size problem.

≥ 1.35 rich

Iron condor

Both shorts placed OUTSIDE the implied move. The edge is the overpricing itself.

0.80 – 1.35 fair

No trade

There is no measurable mispricing here. This band is where the book used to do ALL of its trading.

≤ 0.80 cheap

Debit vertical

In the called direction. The edge is the underpricing.

implied move ÷ realised reaction · thresholds sit well clear of 1.0 because the ratio is computed off four prints — a name at 1.10 is inside its own sampling error

Strikes by distance, not delta

```
shorts at  spot ± 1.0 × implied move
```

The front-weekly surface across a print is too deformed for delta to stand in for distance: a 16-delta strike lands inside or outside the priced move depending only on how the wings are skewed. Where the shorts sit relative to that move IS the thesis — so a coarse strike ladder that snaps one back inside is declined, not quietly downgraded.

The direction call is now a tilt, not the thesis

It picks the side of the debit vertical, and on the condor it pushes the threatened short further out — never pulls the other one in, because collecting more premium by moving a short inside the implied move surrenders the one property the structure is built on.

The first honest backtest made the point in one trade: 22 ideas, 1 approved, full defined risk lost on DG. The structure was doing exactly what it was built to do.

ARCHITECTURE

One pipeline. Nothing but the risk engine can approve.

The live agent and the replay run the same ordered stages, from the same objects. A backtest that skips a stage is a backtest of a different system.

1 Strategy gates

Returns a TradeIdea, or a rejection carrying the gate and the number that stopped it.

2 Modelled cost

Full round-trip estimate attached before anything judges it — so even a rejection carries what it would have cost.

3 Critic

Scores and writes reasoning. It can decline. It cannot approve.

4 Risk engine

The only approver. Fifteen deterministic checks, then a signed stamp.

5 Partner veto

Seven hook points. Veto only — nothing a partner returns can turn a rejection into an approval.

6 Execution

Atomic combo order, deterministic client_order_id, pre-submit existence check so a retry cannot double-fill.

Sizing is from max loss, not from capital.

A structure without a computable max loss is refused outright — which is also what makes a stray undefined-risk idea impossible to execute by accident.

8,569 ideas generated → 441 approved · 5.1%

Fifteen checks, in a fixed order

Hard, deterministic limits enforced in Python before any order is routed. No model can talk its way past them.

`firewall`	`halted`	`market_open`	`undefined_risk`	`unknown_risk`
`leg_count`	`duplicate_legs`	`max_positions`	`max_new_per_day`	`duplicate_structure`
`reentry_cooldown`	`concentration`	`sizing`	`portfolio_risk`	`cash`

The two highlighted checks exist because brokers net identical option symbols: opening the same condor twice does not create a second position, it doubles the quantity on the same four contracts — so every count-based limit was blind to it. Nearly invisible at four cycles a day; severe at twelve.

The capital firewall

15:15 Transient books liquidated and polled until confirmed flat. Carry is untouched.

15:45 Fresh Reg T poll: zero transient exposure, carry margin covered with headroom.

Failure Transient books disabled for the following session. A book that needed rescuing does not get fresh leverage.

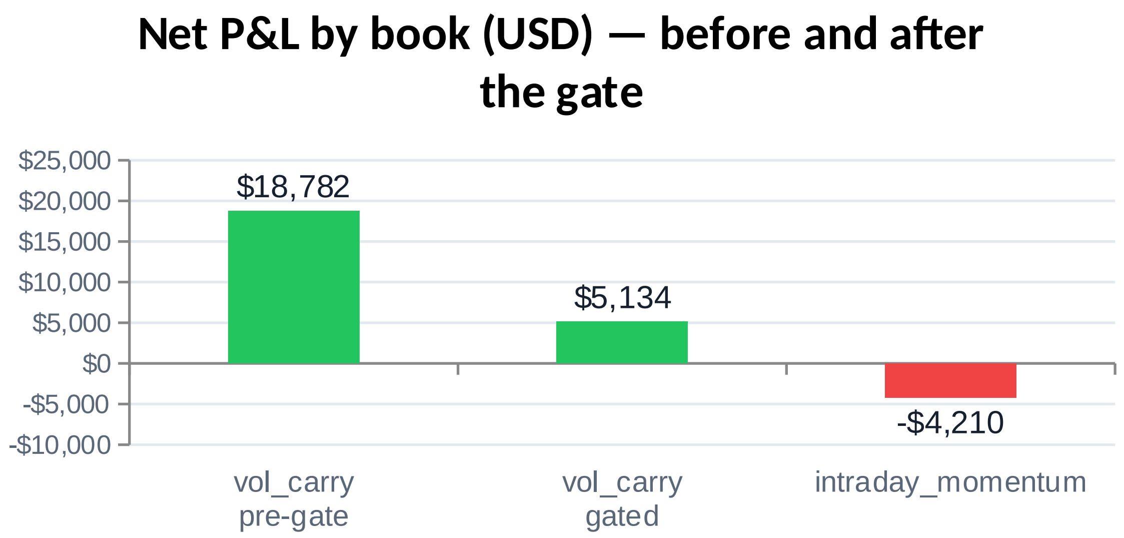
Judged account limits

Risk per trade	**1.5% of equity**
Positions / underlying	**1**
Daily loss halt	**3%**
Net delta cap	**5.5× (observed max 5.24)**
Net vega cap	**$30 per vol pt (obs. 27)**

EVIDENCE · REAL ALPACA BARS, FEB–AUG 2026

We audited our own headline away

409 closed trades after the credit-to-width gate, 14 index and sector ETFs, spread and fees charged inside every fill. The pre-gate run is shown beside it.



+$924
combined net P&L, post-gate

47
impossible condors found

−3.28%
max drawdown

1.03
profit factor

46.7%
win rate

71%
of carry P&L was artefact

Read the artefact number first.

Per-leg implied vols are recovered by inverting Black-Scholes on each leg's own print, with no cross-strike consistency check. A non-monotone smile let the engine book condors collecting more credit than their own width — structures that cannot exist. 47 of 198 carry trades, +$13,249. `risk.max_credit_to_width: 0.45` removes them and the headline falls from +$13,765 to +$924.

vol_carry

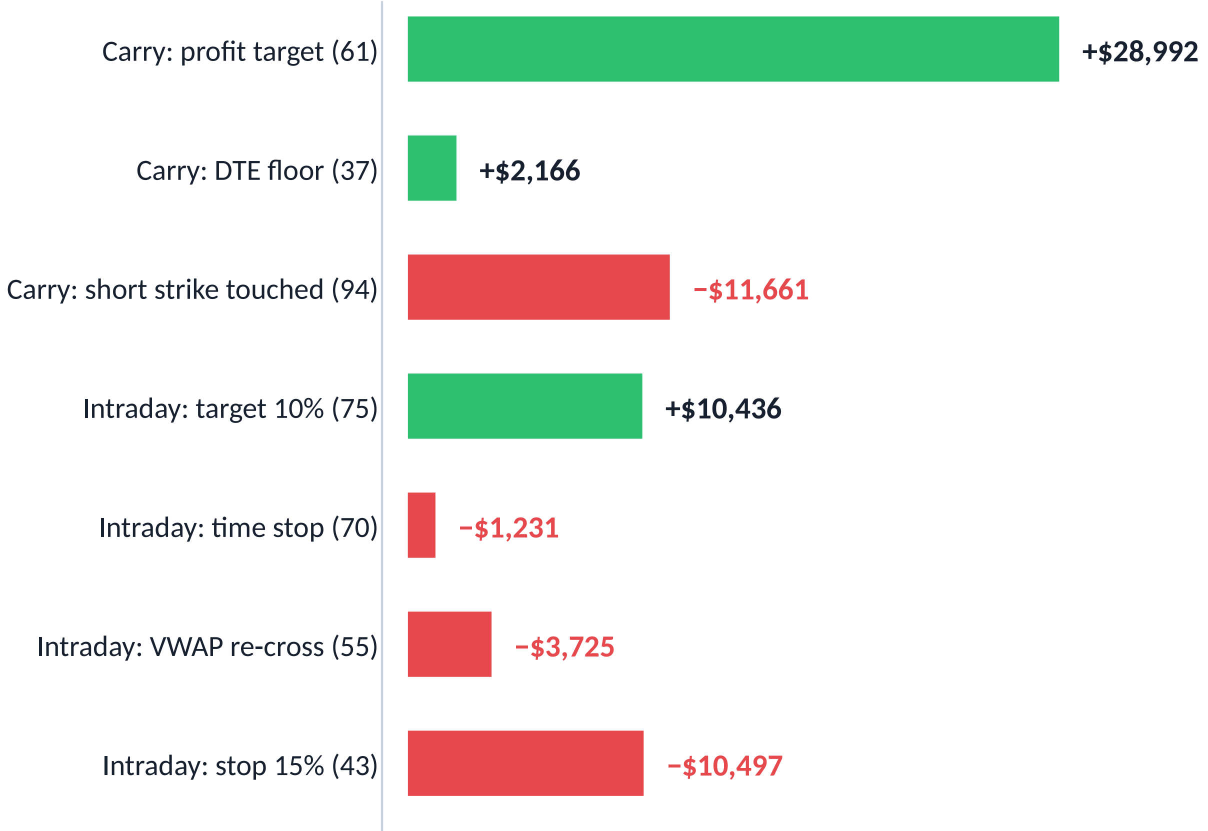
160 trades · 55% win · +$5,134 (pre-gate: +$18,782)

intraday_momentum

249 trades · 41% win · −$4,210

The exits are the strategy

Every closed trade in the pre-gate run, grouped by the rule that closed it — net P&L in USD, trade count in brackets. Entry gates choose the trade; the exit pair decides what it is worth. The credit-to-width gate removes 36 of the carry profit targets; the shape below survives it, the totals do not.



Short-strike touches are the carry book's cost of doing business

94 defensive closes for −$11,661, against 61 profit targets for +$28,992. Closing the tested side early is what keeps a defined-risk structure defined.

The intraday stop is wider than its target on purpose

75 targets at +$139 average against 43 stops at −$244. Premium is noisy and a tight stop is hit by spread flicker alone — but it means the book needs a ~60% hit rate, and it is not getting one.

The VWAP re-cross exit was removed — 30 Aug

55 exits, −$3,725, and 0 wins in 55 attempts. Released, 5 more reached the target, 15 became stops and 35 became time stops. Removed on the mechanism, not the P&L: the improvement is inside noise (t = 0.21).

Carry-only after the gate, Feb–Aug 2026, 14 symbols: +$5,134 · 160 trades · 55.0% win · profit factor 1.375 · t = 1.62. Positive, not significant, and concentrated: the top five trades are 52% of it.

HONESTY

What we know is wrong with it

Every one of these is measured and in the repo. A deck that only lists strengths is a deck that has not been tested.

Exit code is not shared — 29 Aug

Entry is genuinely the same objects live and in replay. Exit is reimplemented in the backtest engine, and nothing asserts the two agree — including a known live defect that compares P&L per leg and can close one leg of a condor on its own.

The intraday signal has no edge after costs — 30 Aug

243 trades, −$5,018, −$20.7 per trade, t = −2.09. No exit change is individually significant, so it is the signal, not the exits. Dropping the book is the biggest weekly-consistency lever: weekly sd 590 → 473, winning weeks 52% → 59%.

Paper fills flatter

Mid fills, no queue, no partials — and spread cost is the primary loss mechanism, which is exactly what paper does not simulate.

Five sessions is not a sample

Expect 10–20 trades in the judged window. That cannot distinguish edge from luck in either direction, and no result from it should be read as if it could.

The live chain was empty — found 31 Aug

Both live providers requested a 3–45 DTE chain while the intraday book selects 0–2, so it reported “no contracts survived the liquidity filter” every cycle — an empty shelf, not a liquidity problem. Deriving the window from the enabled books takes SPY 0–2 DTE contracts from 0 to 324. Fixed on a branch.

The tail is correlated — 30 Aug

The cap is per underlying and blind to correlation. Ten symbols measured as ~2.4 independent bets; the daily loss halt catches the next day, not the same one.

The agent's job is to decline.

8,569 ideas were generated and 409 were taken after the credit-to-width gate. The interesting artefact is not the equity curve — it is the rejection log: every declined trade recorded with the gate that stopped it and the number it measured.

8,569
ideas generated

409
trades taken, post-gate

72,753
rejections logged with a reason

- **Gates are hard, not scored**
  A blended score lets a rich IV reading paper over an earnings date.
- **No model can approve**
  The critic and the partners may only decline. One deterministic engine signs.
- **Every number is defended**
  Each threshold in the config carries the measurement that set it — and the ones that are still guesses say so.

Appendix follows: full parameter sets for all three books, the gate funnel, and the universe rule.

vol_carry — every parameter

ENTRY GATES

iv_rank_min	0.35	Regime filter. Was 0.70, which rejected 304 of 304 replayed candidates.
iv_rv_spread_min	0.03	3 vol points. This is the gate carrying the actual edge.
adx_max / trend	25 / 0.60	Short premium is short movement.
exclude_earnings	true	IV is elevated because of the event.

STRUCTURE

short_delta_target	0.25	Credit-to-width ~30% vs 17% at 14Δ. Hypothesis under measurement.
wing_width_pct	0.015	Fraction of spot, so it means the same on TLT as on SPY.
dte_min / dte_max	7 / 14	At 10 DTE a five-session hold captures most of the remaining decay.
min_credit_to_width	0.15	Below this the credit is not paying for the tail.

EXITS AND COST

profit_target_pct	0.50	With the 1.5× stop this sets breakeven at 75%.
max_loss_usd	450	Was 900 and never fired — the six largest losses were $780–868.
reentry_cooldown	1440 min	One entry per underlying per session for this book.
max_spread_cost	0.20	Of credit received. A condor crosses eight half-spreads round trip.

Universe: SPY QQQ IWM DIA XLF XLE TLT GLD XLV XLU EEM EFA FXI SLV — chosen on measured spread and independence, not on P&L.

intraday_momentum — every parameter

SIGNAL

`band_dispersion_mult`	2.0	Swept on real bars; below 1.5 the band was still the wall.
`cross_lookback_bars`	15	On 1-minute bars = exactly one polling gap.
`volume_zscore_min`	0.5	Top ~31% of that time-of-day bucket. Was 1.0.
`confirmations_required`	3	Of up to seven votes. needed = min(3, possible).
`term_structure band`	−0.10 to 0.25	Requires measured prints; the modelled chain's slope is a constant.

COST AND TIMING

`max_relative_spread`	0.04	2% was below the tightest quote in the universe — unpassable.
`spread_cost / target`	0.30	The binding, economically meaningful test.
`entry window`	09:45–14:45	Lunch 11:30–13:30 skipped.
`dte_max`	2	Front-expiry gamma is the point.

SELECTION AND EXITS

`iv_rank_no_trade_above`	0.85	Declining when the move is already priced in.
`max_option_price`	$25.00	Stops selection landing on a deep-ITM contract.
`target / stop`	10% / 15%	Stop wider on purpose; implies a ~60% breakeven hit rate.
`time_stop / flat_by`	20 min / 15:10	Flat five minutes ahead of the firewall cutoff.

Universe: SPY QQQ IWM DIA XLF XLE TLT GLD — index products only; a $0.10-wide single-name quote costs $20 round trip against a $10–30 target.

earnings_event_directional — every parameter

SCREEN AND CALENDAR

min_implied_move_pct	3.0	Below this no event is priced in, and the round trip still costs four spreads.
max_relative_spread	0.25	The binding gate. GTLB at $45 pays ~$80 of a ~$120 credit to the market maker.
top_n	10	One LLM call each; ten positions sharing one night's risk budget.
calendar confirmation	required	The model proposes; the calendar file confirms. Unverified names never arm.

WATCH AND DIRECTION

lookahead_days	3	Hourly polls from three days out. No new items means no model call.
min_salience / max_notes	0.35 / 40	Ten low-salience notes dilute the two that matter.
min_confidence	0.55	The single most important number in the file — and it sets the size.
expected_abstention	0.40	A zero abstention rate logs a warning: that model is not filtering.

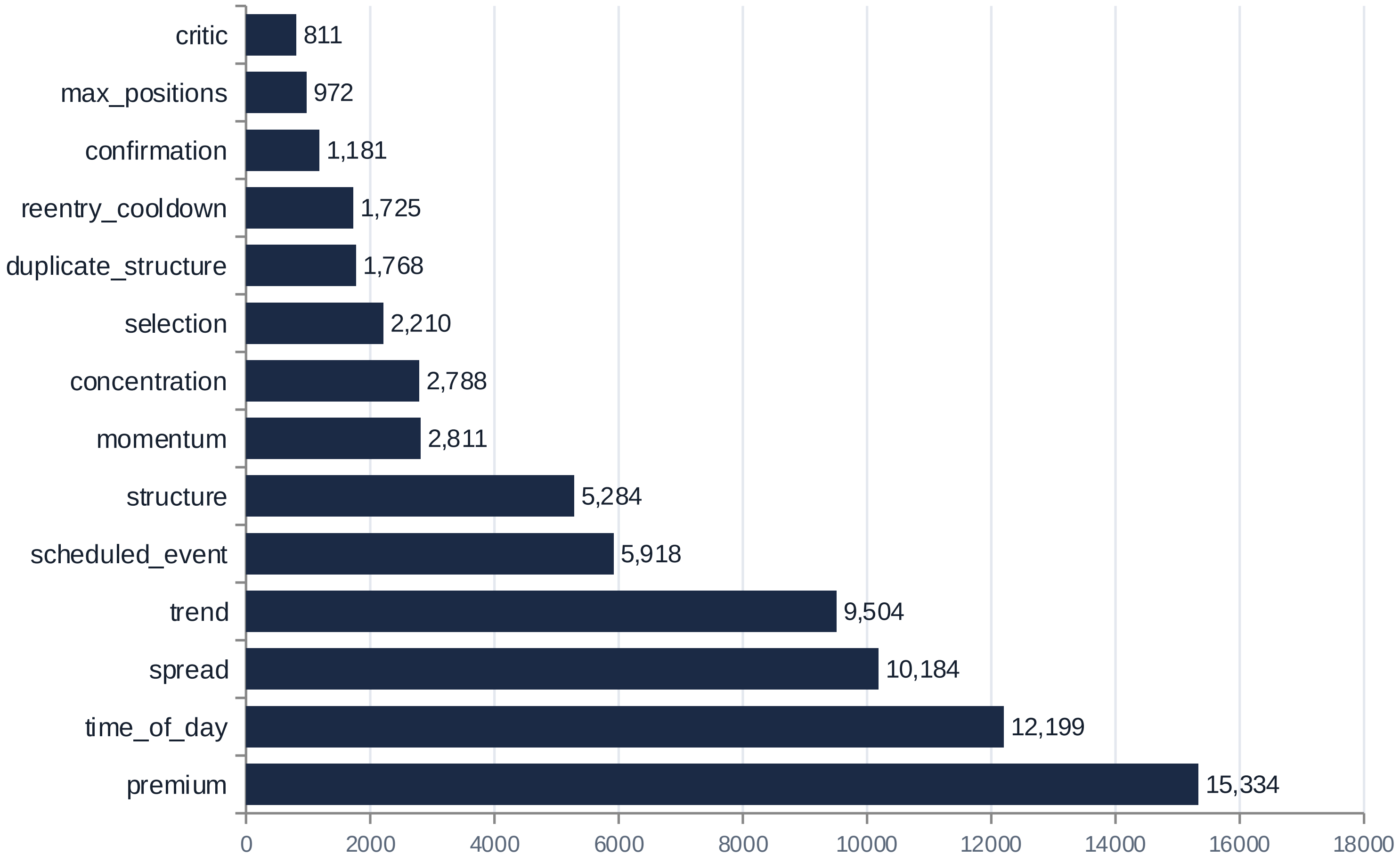
STRUCTURE AND SIZE

rich / cheap threshold	1.35 / 0.80	Outside these, the expression follows the sign. Between them, no trade.
shorts_at_implied_move	1.0	Distance, not delta. min_shorts_clearance 0.85 declines a coarse ladder.
nightly_risk_budget_pct	0.04	Shared across every name opened that night; 1.2% per name.
arm / flatten	15:45 / 09:45	One overnight hold. No intraday management, by design.

Not a firewall tenant: a book whose entire life is one overnight hold does not fit the intraday/carry tenancy model, so it is bounded by size instead.

The rejection funnel

72,753 rejections across the Feb–Aug run, by the gate that stopped the candidate. This log — not the equity curve — is the artefact that shows an agent declining trades with the number that caused it.



`premium`

The carry book's regime filter, and the largest single rejector. It is a filter working, not a block: entries occurred on 43% of sessions.

`spread`

Third largest, and the one that should be large. Execution cost is the binding constraint on both books.

`duplicate_structure + reentry_cooldown`

3,493 rejections that did not exist before 27 Aug, when the book was silently doubling positions instead of opening new ones.

Read it with `oaa backtest --why 15`, which groups by reason and normalises the numbers inside each one.

The universe is a rule, not a list

Three conditions, applied mechanically, so the list follows the measurement rather than the last backtest.

1

Median option spread ≤ 2% of mid

Execution cost decides whether an edge survives at all.

2

It must raise the number of effective independent bets

Not the ticker count. Ten symbols measured as ~2.4 independent bets.

3

No idiosyncratic event risk in a short-vol book

The event gate excludes scheduled earnings. Nothing protects a short condor from a surprise product headline.

What the rule removed, and what it kept

	Chain spread	Corr vs QQQ	Decision
XLK	27.94%	0.96	REMOVED — worst on both criteria at once
SMH	12.49%	0.91	REMOVED — same argument
QQQ	2.85%	–	KEPT — it is what XLK was duplicating
XLE	11.79%	−0.18 to −0.36	KEPT despite the spread — the only negative correlation

A short-premium book's entire failure mode is every position losing at once, so a negatively correlated name is worth more than three more equity ETFs — even an expensive one.